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Design and Implementation of a Buck Converter for DC Voltage Regulation

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Abstract

Abbreviations and Acronyms

API Application Programming Interface

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1 Introduction and Functionality

2 System Architecture

2.1 Triangular Waveform Generation

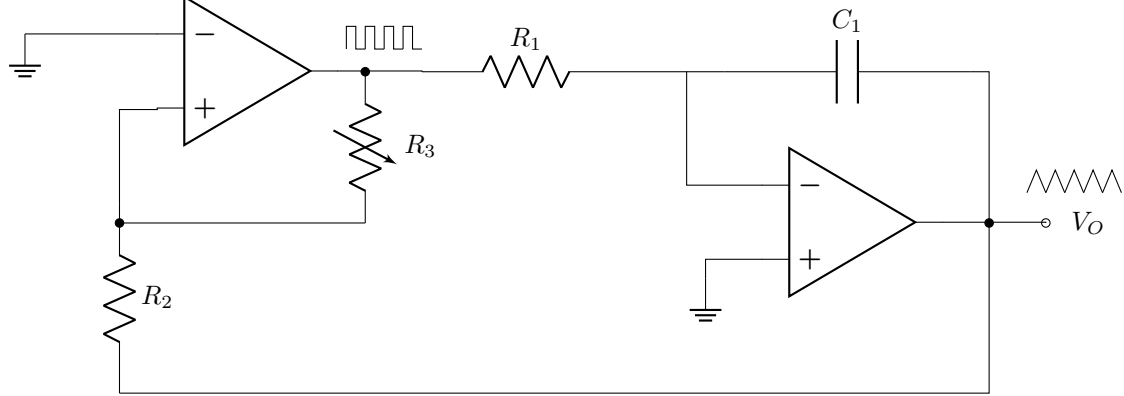


Figure 1: Circuit Diagram of Triangular Waveform Generator

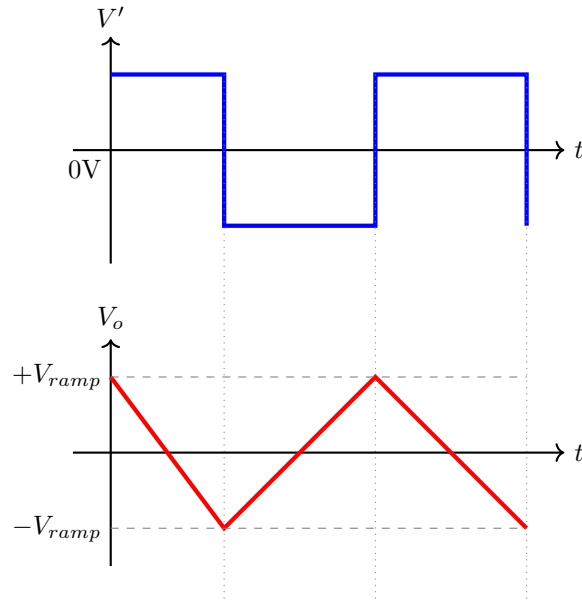


Figure 2: Output waveforms of the Triangular Wave Generator.

2.1.1 Design for Required Specifications

Given: $f = 20$ kHz and $V_{o(pp)} = 4$ V (For Triangular Waves).

Assume the op-amp saturation voltage as $V_{sat} = \pm 7.5$ V.

For amplitude,

$$V_{o(pp)} = 2 \frac{R_2}{R_3} V_{sat}$$

Let's take R_3 as $10\text{ k}\Omega$ and substituting values,

$$4 = 2 \frac{R_2}{10} \times 7.5$$

Therefore, $R_2 = 2.667\text{ k}\Omega$.

Choose standard values as $R_2 = 2.7\text{ k}\Omega$

For frequency,

$$f = \frac{R_3}{4R_1CR_2}$$

Substituting known values,

$$20 \times 10^3 = \frac{10 \times 10^3}{4 \cdot R_1 \cdot C \cdot 2.7 \times 10^3}$$

$$R_1 C = 4.6296 \times 10^{-5}$$

Choose $C = 10\text{ nF}$.

Then,

$$R_1 = \frac{4.6296 \times 10^{-5}}{10 \times 10^{-9}} = 4629.63\ \Omega$$

Choose the nearest standard value: $R_1 = 4.7\text{ k}\Omega$.

Final component values are:

$R_1 = 4.7\text{ k}\Omega$, $R_2 = 2.7\text{ k}\Omega$, $R_3 = 10\text{ k}\Omega$, and $C = 10\text{ nF}$.

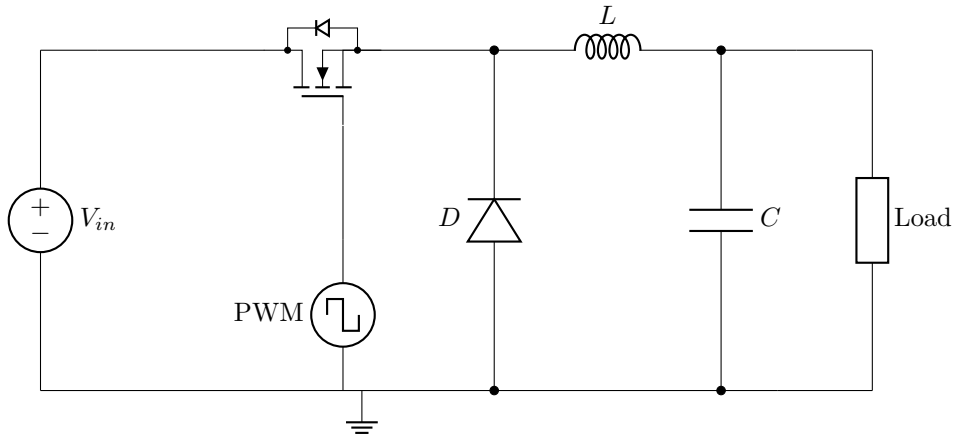


Figure 3: Circuit Diagram

2.2 Buck Converter Design

Buck converter equations relate input voltage (V_{in}), output voltage (V_{out}), duty cycle (D), switching frequency (f_s), and component values (L , C) to achieve step-down DC-DC conversion. These equations are typically derived assuming Continuous Conduction Mode (CCM).

2.2.1 Calculations

Target Inductor Ripple Current (ΔI_L): As a rule of thumb, we can choose 10% of output current.

$$\Delta I_L = 10\% \times I_{out} = 0.1 \times 1 = 0.1\text{ A}$$

Target Inductor Ripple Voltage Also, as a rule of thumb, we can take the Voltage Ripple (ΔV_{out}) as 1% of output voltage.

$$\Delta V_{out} = 1\% \times 12 = 0.12 \text{ V}$$

Duty Cycle Calculation:

$$D = \frac{V_{out}}{V_{in}}$$

$$D_{min} = \frac{3}{15} = 0.2, \quad D_{max} = \frac{12}{15} = 0.8$$

Inductor Calculation (worst case at $V_{out} = 12 \text{ V}$):

$$L = \frac{V_{out}(1 - D)}{f_s \cdot \Delta I_L}$$

$$L = \frac{12(1 - 0.8)}{15000 \times 0.1} = \frac{12 \times 0.2}{1500} = 0.0016 \text{ H} = 1.6 \text{ mH}$$

Capacitor Calculation:

$$C = \frac{\Delta I_L}{8f_s \cdot \Delta V_{out}}$$

$$C = \frac{0.1}{8 \times 15000 \times 0.03} = 27.7778 \mu\text{F}$$

Peak Inductor Current:

$$I_{L,peak} = I_{out} + \frac{\Delta I_L}{2} = 1 + \frac{0.1}{2} = 1.05 \text{ A}$$

Note

All formulas above assume an ideal buck converter (no losses, ideal components and operation in Continuous Conduction Mode).

3 Component Selection

4 Software Simulation and Hardware Testing

5 Conclusion & Future Works

6 Contribution of Group Members

Acknowledgment

References

- [1] EEeguide, "Triangular Wave Generator using Op-amp," [Online]. Available: <https://www.eeeguide.com/triangular-wave-generator-using-op-amp/>.
- [2] Swagatam, "Calculating Inductor Value in SMPS," *Homemade Circuit Projects*, [Online]. Available: <https://www.homemade-circuits.com/calculating-inductor-value-in-smps/>.